

Hussam Amrouch

professor

基本信息

姓名: Hussam Amrouch

所属院校: 斯图加特大学

地区: Germany

年龄范围: 30岁至40岁

海外经历: 有

学科领域

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

人工智能

研究领域

低功耗电路与能量管理

低功耗设计与能耗优化

微电子器件与制造

计算机体系结构

集成电路设计与制造

深度学习与神经网络

EDA工具与电路验证

数字集成电路设计

存储器设计与架构

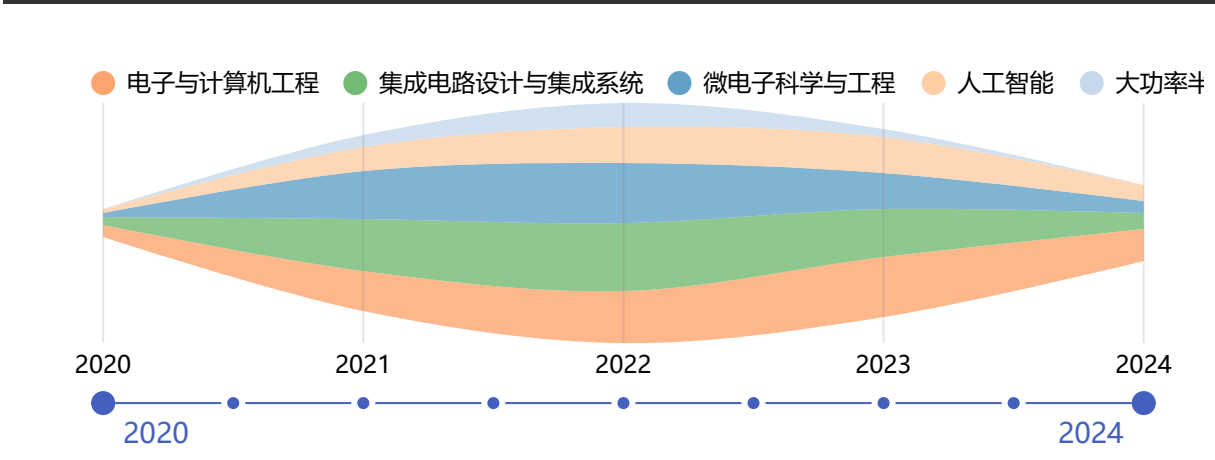
嵌入式系统与硬件设计

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个人简介

Hussam Amrouch is a Prof. Dr.-Ing. He heads the Chair of AI Processor Design at the Technical University of Munich and the Munich Institute of Robotics and Machine Intelligence. He also heads the Semiconductor Test and Reliability at the University of Stuttgart. He received his Ph.D. from KIT in 2015. His research interests include reliability design, machine learning for CAD, etc. He holds eight HiPEAC Paper Awards and has over 200 publications. He serves as an editor at Nature Scientific Reports and is on the committees of major EDA conferences. Contact: +49 (89) 289 - 51410, amrouch(at)tum.de, <https://www.ce.cit.tum.de/aipro/>.

近年发表的论文



Temperature- and variability-aware compact modeling of ferroelectric FDSOI FET for memory and emerging applications
SOLID-STATE ELECTRONICS
Chatterjee, Swetaki; Kumar, Shubham; Gaidhane, Amol; Dabhi, Chetan Kumar; Chauhan, Yogesh Singh; Amrouch, Hussam
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Integrated CPU Monitoring Using 2D Temperature Sensor Arrays Directly Printed on Heat Sinks
ADVANCED MATERIALS TECHNOLOGIES
Huber, Robert; Belles, Daniel; Buecher, Tim; Franke, Leonard; Amrouch, Hussam; Lemmer, Uli
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Introduction to Special Issue on In/Near Memory and Storage Computing for Embedded Systems
ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS
Shi, Liang; Hu, Jingtong; Amrouch, Hussam; Chen, Kuan-Hsun; Zhao, Mengying; Liu, Weichen
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Introduction to the Special Issue on Design for Testability and Reliability of Security-aware Hardware
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Ni, Tianming; Wen, Xiaoqing; Amrouch, Hussam; Zhuo, Cheng; Song, Peilin
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Graph Attention Networks to Identify the Impact of Transistor Degradation on Circuit Reliability
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Mohamed, Tarek; van Santen, Victor M.; Alrahis, Lilas; Sinanoglu, Ozgur; Amrouch, Hussam
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In-Memory Acceleration of Hyperdimensional Genome Matching on Unreliable Emerging Technologies

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Barkam, Hamza E.; Yun, Sanggeon; Genssler, Paul R.; Liu, Che-Kai; Zou, Zhuowen; Amrouch, Hussam; Imani, Mohsen

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WaSSaBi: Wafer Selection With Self-Supervised Representations and Brain-Inspired Active Learning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Pandaram, Karthik; Genssler, Paul R.; Amrouch, Hussam

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DropHD: Technology/Algorithm Co-design for Reliable Energy-efficient NVM-based Hyperdimensional Computing under Voltage Scaling

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Genssler, Paul R.; Mayahiniat, Mahta; Thomann, Simon; Tahoori, Mehdi B.; Amrouch, Hussam

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HDCircuit: Brain-inspired HyperDimensional Computing for Circuit Recognition

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Genssler, Paul R.; Alrahist, Lilas; Sinanoglu, Ozgur; Amrouch, Hussam

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Frontiers in Edge AI with RISC-V: Hyperdimensional Computing vs. Quantized Neural Networks

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Machine Learning Unleashes Aging and Self-Heating Effects: From Transistors to Full Processor

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Time and Space Optimized Storage-based BIST under Multiple Voltages and Variations

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On the Severity of Self-Heating in FDSOI at Cryogenic Temperatures: In-depth Analysis from Transistors to Full Processor

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CARAT-A reliability analysis framework for BTI-HCD aging in circuits

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Golden-Free Robust Age Estimation to Triage Recycled ICs

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Transistor Self-Heating-Aware Synthesis for Reliable Digital Circuit Designs

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Modeling and Predicting Transistor Aging Under Workload Dependency Using Machine Learning

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Compact CMOS-Compatible Majority Gate Using Body Biasing in FDSOI Technology

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Cryogenic In-Memory Computing for Quantum Processors Using Commercial 5-nm FinFETs

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Learning-Oriented Reliability Improvement of Computing Systems From Transistor to Application Level

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Reliable Hyperdimensional Reasoning on Unreliable Emerging Technologies

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Compact and High-Performance TCAM Based on Scaled Double-Gate FeFETs

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Design Automation for Cryogenic CMOS Circuits

Design Automation Conference

Victor M. van Santen; Marcel Walter; Florian Klemme; Shivendra Singh Parihar; Girish Pahwa; Yogesh S. Chauhan; Robert Wille; Hussam Amrouch

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MLCAD: A Survey of Research in Machine Learning for CAD Keynote Paper

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2022

Characterizing Approximate Adders and Multipliers for Mitigating Aging and Temperature Degradations

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2022

Energy Efficient Edge Computing Enabled by Satisfaction Games and Approximate Computing

IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING

Irtija, Nafis; Anagnostopoulos, Iraklis; Zervakis, Georgios; Tsiropoulou, Eirini Eleni; Amrouch, Hussam; Henkel, Joerg

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2022

Brain-Inspired Computing for Circuit Reliability Characterization

IEEE TRANSACTIONS ON COMPUTERS

Genssler, Paul R. R.; Amrouch, Hussam

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2022

Trojan Detection in Embedded Systems With FinFET Technology

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Surabhi, Virinchi Roy; Krishnamurthy, Prashanth; Amrouch, Hussam; Henkel, Joerg; Karri, Ramesh; Khorrami, Farshad

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2022

Thermal-Aware Design for Approximate DNN Accelerators

IEEE TRANSACTIONS ON COMPUTERS

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2022

FeFET-Based Binarized Neural Networks Under Temperature-Dependent Bit Errors

IEEE TRANSACTIONS ON COMPUTERS

Yayla, Mikail; Buschjaeger, Sebastian; Gupta, Aniket; Chen, Jian-Jia; Henkel, Joerg; Morik, Katharina; Chen, Kuan-Hsun; Amrouch, Hussam

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2022

Real-Time Full-Chip Thermal Tracking: A Post-Silicon, Machine Learning Perspective

IEEE TRANSACTIONS ON COMPUTERS

Sadiqbatcha, Sheriff; Zhang, Jinwei; Amrouch, Hussam; Tan, Sheldon X. -D.

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2022

Impact of NCFET Technology on Eliminating the Cooling Cost and Boosting the Efficiency of Google TPU

IEEE TRANSACTIONS ON COMPUTERS

Salamin, Sami; Zervakis, Georgios; Klemme, Florian; Kattan, Hammam; Chauhan, Yogesh; Henkel, Joerg; Amrouch, Hussam

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2022

On the Reliability of FeFET On-Chip Memory

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2022

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Paim, Guilherme; Amrouch, Hussam; Rocha, Leandro M. G.; Abreu, Brunno; Cesar da Costa, Eduardo Antonio; Bampi, Sergio; Henkel, Joerg

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Alahis, Lilas; Knechtel, Johann; Klemme, Florian; Amrouch, Hussam; Sinanoglu, Ozgur

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2022

A Novel Attack Mode on Advanced Technology Nodes Exploiting Transistor Self-Heating

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2022

Electrothermal Simulation and Optimal Design of Thermoelectric Cooler Using Analytical Approach

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Full-Chip Power Density and Thermal Map Characterization for Commercial Microprocessors Under Heat Sink Cooling

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2022

Bridging the Gap Between Voltage Over-Scaling and Joint Hardware Accelerator-Algorithm Closed-Loop

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Paim, Guilherme; Amrouch, Hussam; da Costa, Eduardo Antonio Cesar; Bampi, Sergio; Henkel, Joerg

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Efficient Learning Strategies for Machine Learning-Based Characterization of Aging-Aware Cell Libraries

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Variability-Aware Approximate Circuit Synthesis via Genetic Optimization

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2022

Reliable Binarized Neural Networks on Unreliable Beyond Von-Neumann Architecture

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2022

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Ravipati, Divya Praneetha; Kedia, Rajesh; van Santen, Victor M.; Henkel, Jorg; Panda, Preeti Ranjan; Amrouch, Hussam

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All-in-Memory Brain-Inspired Computing Using FeFET Synapses

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Leveraging Ferroelectric Stochasticity and In-Memory Computing for DNN IP Obfuscation

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Mankali, Likhitha; Rangarajan, Nikhil; Chatterjee, Swetaki; Kumar, Shubham; Chauhan, Yogesh Singh; Sinanoglu, Ozgur; Amrouch, Hussam

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Design Close to the Edge for Advanced Technology using Machine Learning and Brain-Inspired Algorithms

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AppGNN: Approximation-Aware Functional Reverse Engineering using Graph Neural Networks

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Compact Ferroelectric Programmable Majority Gate for Compute-in-Memory Applications

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Novel FDSOI-based Dynamic XNOR Logic for Ultra-Dense Highly-Efficient Computing

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Cleaved-Gate Ferroelectric FET for Reliable Multi-Level Cell Storage

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Bagga, Navjeet; Ni, Kai; Chauhan, Nitanshu; Prakash, Om; Hu, X. Sharon; Amrouch, Hussam

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Ferroelectric FET Threshold Voltage Optimization for Reliable In-Memory Computing

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Prakash, Om; Ni, Kai; Amrouch, Hussam

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Suppressing Channel Percolation in Ferroelectric FET for Reliable Neuromorphic Applications

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Wafer Map Defect Classification Based on the Fusion of Pattern and Pixel Information

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Liao, Yiwen; Latty, Raphael; Genssler, Paul R.; Amrouch, Hussam; Yang, Bin

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Joint Modeling of Multi-Domain Ferroelectric and Distributed Channel towards Unveiling the Asymmetric Abrupt DC Current Jump in Ferroelectric FET

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Brain-Inspired Hyperdimensional Computing for Ultra-Efficient Edge AI

2022 INTERNATIONAL CONFERENCE ON HARDWARE/SOFTWARE CODESIGN AND SYSTEM SYNTHESIS (CODES+ISSS)

Amrouch, Hussam; Imani, Mohsen; Jiao, Xun; Aloimonos, Yiannis; Fermuller, Cornelia; Yuan, Dehao; Ma, Dongning; Barkam, Hamza E.; Genssler, Paul R.; Sutor, Peter

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HW/SW Codesign for Approximate In-Memory Computing

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Power-Efficient Heterogeneous Many-Core Design With NCFET Technology

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On-Demand Mobile CPU Cooling With Thin-Film Thermoelectric Array	
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Automated Design Approximation to Overcome Circuit Aging	
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Variability Effects in FinFET Transistors and Emerging NC-FinFET

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Towards Reliable In-Memory Computing: From Emerging Devices to Post-von-Neumann Architectures

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Klemme, Florian; Amrouch, Hussam

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On the Resiliency of NC-FinFET SRAMs against Variation: MFIS Structure

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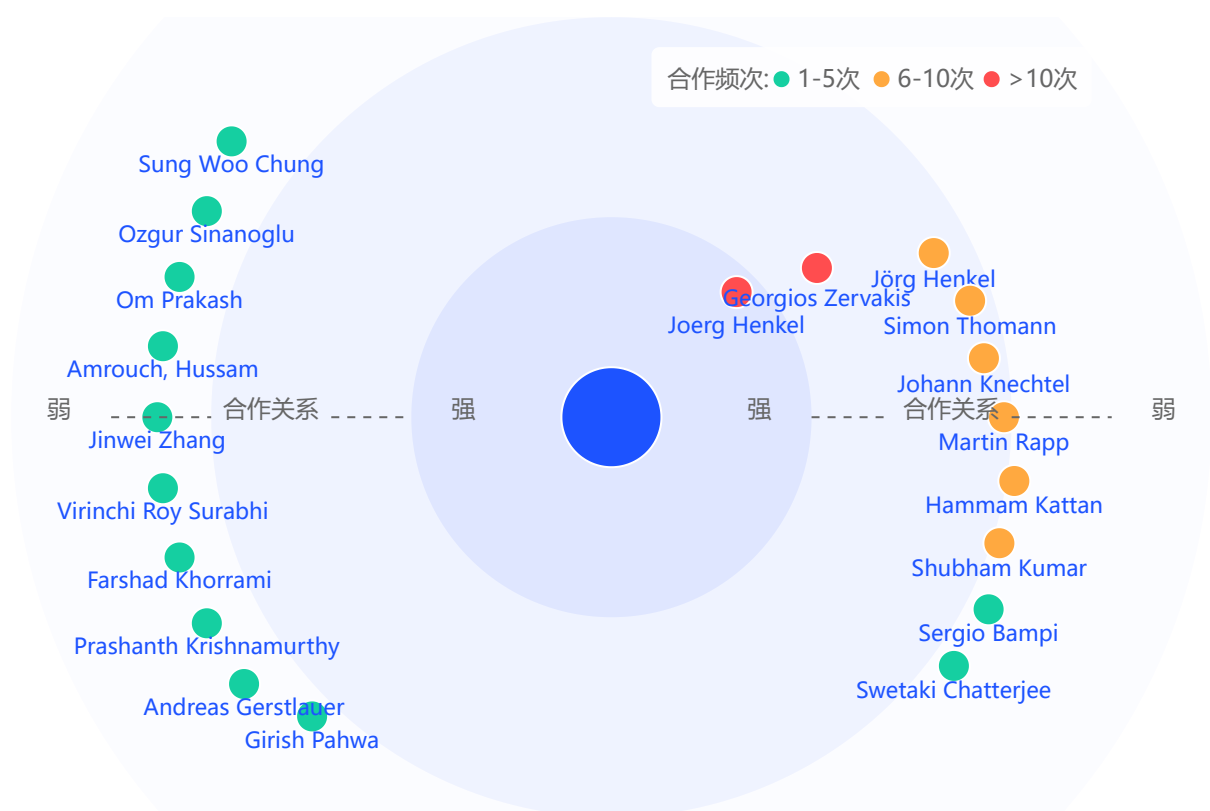
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